S&DS 6150: Introduction to random matrix theory + applications

(1) Covariance estimation: $X_1, \dots, X_n \in \mathbb{R}^d$ i.i.d

$\mathbb{E} X_i = 0, \quad \mathbb{E} X_i X_i^T = \Sigma$

Sample covariance: $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i^T$.

How accurate is this estimate: $\|\hat{\Sigma} - \Sigma\|_{op}$?

Is there a <u>better</u> estimator? Optimal estimator?

(2) PCA: $X_1, \dots, X_n \in \mathbb{R}^d$ i.i.d, $\mathbb{E} X_i = 0$, $\mathbb{E} X_i X_i^T = \Sigma$

1st principal component $\leftrightarrow$ leading eigenvector of $\hat{\Sigma}$

(direction in $\mathbb{R}^d$ of largest data variation)

Spiked model (Johnstone '01): $\Sigma = I + \lambda v v^T$.

How accurate is PCA: $|\hat{v}^T v|$?

(3) Ridge regression: $(X_1, Y_1), \dots (X_n, Y_n) \in \mathbb{R}^d \times \mathbb{R}$ i.i.d

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^d}{\operatorname{argmin}} \frac{1}{2} \sum_{i=1}^n (Y_i - X_i^T \beta)^2 + \frac{\lambda}{2} \|\beta\|_2^2$$

Minimum-norm interpolant ($d > n$): $\lambda \to 0$.

Predict generalization error $\mathbb{E}(Y_{test} - X_{test}^T \hat{\beta})^2$?

Why does this generalize in the interpolating regime?

(4) Community detection / spectral clustering: Graph on $\{1, \dots, n\}$.

$A \in \{0,1\}^{n \times n}$ adjacency matrix: $a_{ij} = \begin{cases} 1 & \text{if } i \sim j \\ 0 & \text{if } i \not\sim j \end{cases}$

Spectral clustering: (1) Compute top k eig-vectors $v_1, \dots, v_k$

(2) Perform k-means on rows of $[v_1, \dots, v_k] \in \mathbb{R}^{n \times k}$

Stoch. block model: $\mathbb{P}[i \sim j] = \begin{cases} p & \text{if } i,j \in U \text{ or } i,j \in V \\ q & \text{otherwise.} \end{cases}$

$\mathbb{E}A = \left(\begin{array}{c|c} p & q \\ \hline q & p \end{array}\right)$ (rank 2)

For what (p, q) can this accurately recover U, V?

(5) Spectrum denoising: $X = A + Z \in \mathbb{R}^{n \times n}$ (signal + iid noise)

How do singular values/vectors of X relate to A?

Can we estimate sing. value distribution of A from X?

(6) Expander graphs: $A \in \{0,1\}^{n \times n}$ adj. matrix of d-regular graph

(Largest eig.-value $\lambda_1 = d$)

Alon, Boppana '91: $\max(|\lambda_2|, \dots, |\lambda_n|) \geq 2\sqrt{d-1} - o_n(1)$.

Graphs w/ $\max(|\lambda_2|, \dots, |\lambda_n|) \approx 2\sqrt{d-1}$ are good expanders.

Why do <u>random</u> d-regular graphs achieve this bound?

Goal: Understand — from a non-spectral description of a random matrix — its spectral properties.

Wigner's semicircle law

Def: Let $\{z_{ij}\}_{1 \le i,j \le n}$ be independent (real) r.v.'s.

- $\mathbb{E}\, z_{ij} = 0 \quad \forall i,j$
- $\mathbb{E}\, z_{ij}^2 = 1 \quad \forall i \le j$ (no requirement for $i > j$)
- $\mathbb{E}\, |z_{ij}|^k \le C_k \quad \forall i,j$, each $k \ge 1$, some constant $C_k > 0$.

The symmetric matrix $X \in \mathbb{R}^{n \times n}$ w/

$$x_{ij} = \begin{cases} z_{ij}/\sqrt{n} & i \le j \\ z_{ji}/\sqrt{n} & i > j \end{cases}$$

is a Wigner matrix.

Def: If $z_{ij} \sim \begin{cases} \mathcal{N}(0,1) & i < j \\ \mathcal{N}(0,2) & i = j \end{cases}$ then the law of X is

the Gaussian Orthogonal Ensemble, $X \sim GOE$.

(1) Equivalently, let $W \in \mathbb{R}^{n \times n}$ be asymmetric w/ iid $\mathcal{N}(0,1)$ entries.

$X \sim GOE$ is law of $\frac{W + W^T}{\sqrt{2n}}$. For any orthogonal matrix $O \in \mathbb{R}^{n \times n}$, since $W \overset{L}{=} OWO^T$ also

$$X \overset{L}{=} OXO^T$$

If $X = V\begin{pmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{pmatrix}V^T$ is spectral decomp., $\lambda_1 \geq \dots \geq \lambda_n$,

then $V \overset{d}{=} OV$. I.e., eig.-vectors uniformly distributed on $\mathcal{O}(n)$.

(2) Can extend to complex Hermitian matrices:

z_{ij} real $\forall i,j$, $\mathbb{E}|z_{ij}|^2 = 1$ $\forall i \neq j$; $X_{ij} = \begin{cases} z_{ij}/\sqrt{n} & i<j \\ \overline{z_{ij}}/\sqrt{n} & i>j \end{cases}$

If $W \in \mathbb{C}^{n\times n}$ has i.i.d $\mathcal{N}_{\mathbb{C}}(0,1)$ entries, then

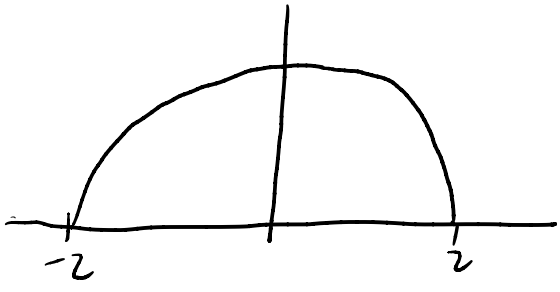
$$X = \frac{W+W^*}{\sqrt{2n}} \sim \text{GUE (Gaussian Unitary Ensemble)}$$

Def: For $X \in \mathbb{R}^{n\times n}$ symmetric, its <u>empirical spectral distribution</u> (ESD) is the discrete probability measure

$$\mu_n = \frac{1}{n}\sum_{i=1}^n \delta_{\lambda_i},$$

$\{\lambda_i\}_{i=1}^n$ the (real) eig.-values of X.

Thm (Wigner): Let $X \in \mathbb{R}^{n\times n}$ be Wigner. As $n\to\infty$, almost surely, its ESD μ_n converges weakly to the <u>semicircle law</u> μ_{sc} on $[-2,2]$, w/ density $f(x) = \frac{1}{2\pi}\sqrt{4-x^2}\,\mathbb{1}_{[-2,2]}(x)$.



Proof strategy: Show $\int f \, d\mu_n \to \int f \, d\mu_{sc}$ for class of $f: \mathbb{R} \to \mathbb{R}$

s.t.

- sufficiently large/rich to imply weak convergence
- $\int f \, d\mu_n = \frac{1}{n} \sum_{i=1}^{n} f(\lambda_i)$ can be described simply via entries of X_n.

Lectures 1+2 (moment method): polynomials generated by $f(x) = x^k$.

$$\int f \, d\mu_n = \frac{1}{n} \sum_{i=1}^{n} \lambda_i^k = \frac{1}{n} \operatorname{Tr} X^k.$$

Lectures 3+4 (Stieltjes transform/resolvent method): rational functions generated by $f(x) = \frac{1}{x - z}$, $z \in \mathbb{C}$.

$$\int f \, d\mu_n = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{\lambda_i - z} = \frac{1}{n} \operatorname{Tr} (X - zI)^{-1}.$$

What are the moments of μ_{sc}?

Prop: Let $m_k = \int x^k \, d\mu_{sc}$. Then $m_{2k} = C_k := \frac{(2k)!}{k!(k+1)!}$, $m_{2k+1} = 0$.

Proof: Induction on k + integration-by-parts

C_k is the kth Catalan number, defined equivalently by

$$C_k = \sum_{i=0}^{k-1} C_i \cdot C_{k-1-i}$$

66 interpretations (Stanley, Enumerative Combinatorics: Vol 2), including

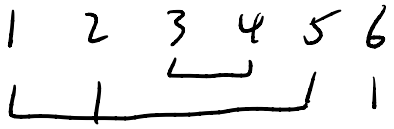
- Number of Dyck words of length $2k$, e.g.

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[Look when first (is paired by):

If inside is Dyck word of length $2i$, then outside is Dyck word of length $2(k-1-i)$. $\Rightarrow C_k = \sum_{i=0}^{k-1} C_i \cdot C_{k-1-i}$]

- Number of non-crossing partitions of $\{1, \dots, k\}$:



$\{1,2,5\}, \{3,4\}, \{6\}$



$\{1,3,5\}, \{2,6\}, \{4\}$

[Look at largest number j in set containing 1:

Removing j induces non-crossing partitions of $\{1, \dots, j-1\}, \{j+1, \dots, k\}$

$\Rightarrow C_k = \sum_{j=1}^{k} C_{j-1} \cdot C_{k-j} = \sum_{i=0}^{k-1} C_i \cdot C_{k-1-i}$]

Lemma: Let X be a Wigner matrix, ESD μ_n. Then $\forall k \geq 1$,

$$\lim_{n \to \infty} \mathbb{E} \int x^k \, d\mu_n = \int x^k \, d\mu_{sc}.$$

Proof: With $\int x^k \, d\mu_n = \frac{1}{n} \sum_{i=1}^{n} \lambda_i^k = \frac{1}{n} \operatorname{Tr} X^k$.

- (i,j) entry of X^2 is $\sum_{k=1}^{n} X_{ik} X_{kj}$
- (i,j) entry of X^3 is $\sum_{l=1}^{n} (X^2)_{il} X_{lj} = \sum_{k,l=1}^{n} X_{ik} X_{kl} X_{lj}$

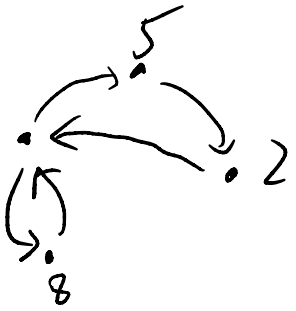
• (i,j) entry of X^k is $\sum_{i_2,\dots,i_k=1}^{n} X_{ii_2} X_{i_2 i_3} \cdots X_{i_k j}$

$$\Rightarrow \frac{1}{n} \operatorname{Tr} X^k = \frac{1}{n} \sum_{i_1,\dots,i_k=1}^{n} X_{i_1 i_2} X_{i_2 i_3} \cdots X_{i_n i_1}$$

Each summand indexed by tuple $(i_1, i_2, \dots, i_k)$.

Associate to a walk on graph on $\{1,\dots,n\}$ w/ edges

$$(i_1, i_2), (i_2, i_3), \dots (i_k, i_1):$$

$(3,5,2,3,8) \longleftrightarrow$ 

Complete undirected graph on $\{1,\dots,n\}$ has $\frac{n(n+1)}{2}$ distinct edges (w/ self-loops), corresponding to independent entries of X.

For a walk $(i_1,\dots,i_k)$, let

$\mathcal{E}$ = set of distinct traversed edges

$k(e)$ = # times $e \in \mathcal{E}$ is traversed

Then by independence,

$$\mathbb{E}\left[X_{i_1 i_2} \cdots X_{i_k i_1}\right] = \mathbb{E} \prod_{e \in \mathcal{E}} X_e^{k(e)} = \prod_{e \in \mathcal{E}} \mathbb{E} X_e^{k(e)}$$

($X_e \stackrel{d}{=} X_{ij}$ if $e = (i,j)$)

Divide walks into 3 cases:

① Some $e \in \mathcal{E}$ is traversed once, i.e. $k(e)=1$.

Then $\mathbb{E} X_e^{k(e)} = 0 \Rightarrow \mathbb{E}[X_{i_1 i_2} \cdots X_{i_k i_1}] = 0$

All other walks traverse

$$\#\text{ distinct edges} \leq \begin{cases} \frac{k}{2} & \text{if } k \text{ even} \\ \frac{k-1}{2} & \text{if } k \text{ odd} \end{cases}$$

Since walk is connected, it traverses

$$\#\text{ distinct vertices} \leq \#\text{ distinct edges} + 1 \leq \frac{k}{2} + 1.$$

② Number of distinct vertices is $< \frac{k}{2}+1$ strictly, i.e. $\leq \frac{k+1}{2}$.

Any such walk can be chosen by:

- First picking $\frac{k+1}{2}$ vertices from $\{1, \ldots, n\}$
- Assigning these vertices to $i_1, \ldots, i_k$.

So # such walks $\leq \binom{n}{\frac{k+1}{2}} \times \left(\frac{k+1}{2}\right)^k \leq C_k \cdot n^{\frac{k+1}{2}}$

for some constant $C_k > 0$.

Recall $X_{ij} = \frac{Z_{ij}}{\sqrt{n}}$ where Z_{ij} has bounded moments of all orders.

Thus, since $\sum_{e \in \mathcal{E}} k(e) = k$, we have $\left| \prod_{e \in \mathcal{E}} \mathbb{E} X_e^{k(e)} \right| \leq \frac{C_k'}{n^{k/2}}$

For some constant $C_k' > 0$

$$\Rightarrow \left| \mathbb{E} \frac{1}{n} \sum_{\substack{\text{case (2) walks} \\ i_1, \dots, i_k}} X_{i_1 i_2} \cdots X_{i_k i_1} \right|$$

$$\leq \frac{1}{n} \times C_k n^{\frac{k+1}{2}} \times \frac{C_k'}{n^{k/2}} \leq \frac{C_k C_k'}{\sqrt{n}}$$

(3) Number of distinct vertices $= \frac{k}{2}+1$. Then

- Number of distinct edges $= \frac{k}{2}$, and k is even
- Graph of distinct traversed edges is a tree, w/ no self-loops.

For each such walk, $k(e) = 2 \ \forall e \in \mathcal{E}$, and $\mathbb{E} X_e^2 = \frac{1}{n}$, so

$$\prod_{e \in \mathcal{E}} \mathbb{E} X_e^{k(e)} = \frac{1}{n^{k/2}}.$$

To count number of such walks:

- Call $(i_1, \dots, i_k)$ and $(i_1', \dots, i_k')$ equivalent if some permutation of $\{1, \dots, n\}$ maps one to the other.

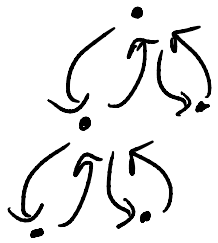
  [E.g. $(3, 5, 2, 3, 8) \leftrightarrow (1, 2, 3, 1, 4)$]

- For each equivalence class (visiting $\frac{k}{2}+1$ vertices), there are $n(n-1)(n-2)\cdots(n-\frac{k}{2})$ ways to assign labels $\{1, \dots, n\}$ to these vertices, i.e. $n(n-1)\cdots(n-\frac{k}{2})$ distinct walks.

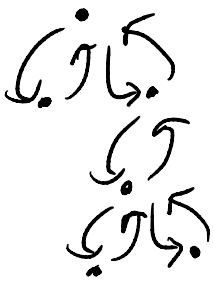
- Since traversed graph is a tree, to specify an equivalence class, we must specify for each of the k steps whether (a) we visit a new vertex, or

(b) we return to its parent

This is in 1-to-1 correspondence w/ Dyck words of length k:



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Thus the number of equivalence classes is $C_{k/2}$.

$$\Rightarrow \mathbb{E}\left[\frac{1}{n}\sum_{\substack{\text{Case ③ walks}\\ i_1 \to \cdots \to i_k}} X_{i_1 i_2} \cdots X_{i_k i_1}\right]$$

$$= \frac{1}{n} \times C_{k/2}\, n(n-1)\cdots\left(n-\tfrac{k}{2}\right) \times \frac{1}{n^{k/2}} = C_{k/2}\,(1+o(1))$$

Thus: $\mathbb{E}\,\frac{1}{n}\operatorname{Tr} X^k = \mathbb{E}\Big[\underbrace{\frac{1}{n}\sum_{\text{Case ①}}}_{=0} + \underbrace{\frac{1}{n}\sum_{\text{Case ②}}}_{\to 0} + \underbrace{\frac{1}{n}\sum_{\text{Case ③}}}_{\begin{cases} =0 \text{ if } k \text{ odd} \\ \to C_{k/2} \text{ if } k \text{ even} \end{cases}}\Big]$

$$\to \int x^k \, d\mu_{sc}$$

Problem 1: (a) Let $A \in \{0,1\}^{n \times n}$ be adj. matrix of Erdos-Renyi graph: $(a_{ij})_{i \le j} \overset{iid}{\sim} \text{Bernoulli}(p_n)$, $a_{ji} := a_{ij}$, $p_n \in (0, 1/2)$.

Let $X = \left(\frac{a_{ij} - p_n}{\sqrt{p_n(1-p_n)}} \right)_{i,j=1}^{n} \in \mathbb{R}^{n \times n}$

Show that, as long as $np_n \to \infty$ as $n \to \infty$,

$$\mathbb{E} \frac{1}{n} \operatorname{Tr} X^k \to \int x^k \, d\mu_{sc}.$$

(b) Let $s : [0,1] \times [0,1] \to [0, \infty)$ be a symmetric, bounded continuous function. Let $\{z_{ij}\}_{i \le j}$ be independent w/

$$\mathbb{E} z_{ij} = 0, \quad \mathbb{E} z_{ij}^2 = s\left(\frac{i}{n}, \frac{j}{n}\right), \quad \mathbb{E}|z_{ij}|^k \le C_k \ \forall k > 0.$$

Let $z_{ji} := z_{ij}$, $X = \left(\frac{z_{ij}}{\sqrt{n}} \right)_{i,j=1}^{n} \in \mathbb{R}^{n \times n}$.

Show that, as long as $\int s(x,y) \, dx = 1 \ \forall y \in [0,1]$,

$$\mathbb{E} \frac{1}{n} \operatorname{Tr} X^k \to \int x^k \, d\mu_{sc}.$$